SPHERE: Spherical-Manifold Equivariant Cross-Modal Foundation Distillation for Universal LiDAR Registration

Anonymous submission

Abstract

Outdoor fleets increasingly deploy mixed LiDAR modalities, yet most learned registration methods remain constrained to a single beam pattern. To address this heterogeneity, we leverage a vision foundation model prior transferred from RGB to LiDAR for cross-sensor registration. Our framework, SPHERE, distills a foundational image backbone to embed semantic information into the 3D representation. This semantic anchoring provides robustness against variations in LiDAR point density and scan patterns, enabling zero-shot generalization across different sensor architectures.

SPHERE operates as a two-stage cross-modal distillation framework utilizing a frozen DINOv2 teacher and a Point Transformer v3 (PTv3) student. Beyond standard contrastive distillation, we introduce two training-time objectives to capture spatial and geometric structures: (i) Spherical-manifold alignment $\mathcal{L}_{\text{sph}}$ transfers the teacher's semantic knowledge by respecting its native hyperspherical geometry. Optimizing true angular distances instead of Euclidean metrics prevents feature distortion and preserves discriminability for point-level matching on sparse scans. (ii) Soft $\mathrm{SE}(3)$ invariance $\mathcal{L}_{\text{eq}}$ enforces viewpoint-independent representations by penalizing descriptor variations across random rigid transformations, minimizing feature drift during deployment.

Evaluated using a single checkpoint trained on nuScenes and KITTI (Stage 1) and fine-tuned on LiDAR-only pairs (Stage 2), SPHERE generalizes robustness across both single-sensor and cross-sensor configurations. The end-to-end framework bridges density and pattern discrepancies between dense training scans and sparse deployment environments through a density-aware training strategy and a lightweight built-in initializer. Crucially, inference operates entirely camera-free, using only a built-in mini-RANSAC initializer and local geometric refinement, without external certifiable solvers or ICP. Under this unified recipe, SPHERE achieves a strict recall (SR@$0.5\,\mathrm{m}/1^{\circ}$) of 97.84% on KITTI, 99.2% on nuScenes, and 95.8% overall on HeLiPR zero-shot cross-sensor pairs, including 94.7% on the sparse 16-beam Velodyne sensor, outperforming BUFFER-X by up to $+41.4$ pp on HeLiPR. On nuScenes and HeLiPR, it matches or exceeds GeoTransformer and BUFFER-X; on KITTI it remains competitive with top geometric matchers while enabling camera-free cross-sensor deployment.

Introduction

Point cloud registration estimates a rigid transform $T \in \mathrm{SE}(3)$ that aligns a source scan P_q to a target P_t. In autonomous driving and robotics, this is a standard component for mapping, localization, and multi-session map updates. In real-world fleet deployments, these updates increasingly involve heterogeneous sensor configurations, where base maps constructed by one sensor type must be incrementally updated using scans from different sensor architectures. This cross-sensor regime introduces domain discrepancies in point densities and scan patterns, limiting the accuracy and robustness of existing registration techniques. Although the task admits a standard evaluation protocol based on recall at translation and rotation thresholds, most existing registration models assume homogeneous sensor setups, leaving the challenges of heterogeneity unaddressed.

Operational heterogeneity manifests through hardware discrepancies and temporal shifts: (a) sensor disparity driven by incompatible vertical elevation profiles, (b) sparsity from legacy 16-beam Velodyne retrofits on auxiliary platforms, (c) non-repetitive Lissajous scan patterns from solid-state Livox Avia units, and (d) FMCW Aeva returns that append radial velocity channels onto raw 3D coordinates. Under these geometric variations, classical descriptors like FPFH (Rusu, Blodow, and Beetz 2009) remain sensor-agnostic but degrade under low overlap or mismatched point densities. Similarly, state-of-the-art learned geometric matchers trained on a single sensor distribution (Qin et al. 2022; Huang et al. 2024) saturate in-distribution but fail when confronted with out-of-distribution beam counts or temporal gaps.

Recent literature addresses this generalization gap. For example, BUFFER-X (Seo et al. 2025) introduces adaptive voxelization and patch-wise scale normalization to achieve zero-shot registration across diverse scenes without manual parameter tuning. Meanwhile, RAP (Pan et al. 2025) casts multi-view registration as a scalable flow-matching problem to align noisy, non-uniform point clouds across domains. However, these approaches remain reliant on geometric cues. When geometry becomes ambiguous due to low sensor resolution or out-of-distribution beam patterns, geometric matchers degrade or require post-processing.

Vision foundation models (VFMs) offer a complementary semantic prior to resolve geometric ambiguities. Self-supervised Vision Transformers (ViTs) such as DINOv2 (Oquab et al. 2024) produce normalized patch features that are stable across varying viewpoints. Recent attempts to leverage these priors in 3D either project the 2D features onto point clouds at test time (Vödisch et al. 2025) or distill them for semantic segmentation tasks (Sautier et al. 2022; Puy, Boulch, and Marlet 2024; Puy et al. 2024). These approaches do not target camera-free outdoor registration across heterogeneous sensors.

To address these limitations, we propose SPHERE, a two-stage cross-modal distillation framework. Instead of modifying geometric heuristics, we reframe the task as cross-modal representation learning, transferring the semantic knowledge of a frozen image foundation network into a 3D LiDAR student. This semantic anchoring enables the representation to generalize across variations in point density and scanning patterns. To ensure deployability, SPHERE introduces geometric and perspective constraints, including a spherical-manifold alignment that preserves the geometry of the vision teacher and a soft $\mathrm{SE}(3)$ invariance constraint that minimizes descriptor drift under viewpoint shifts. By embedding image-based priors into the 3D representation, SPHERE consolidates the registration pipeline into a single student backbone, a correspondence head, and a deep pose pipeline. Our framework achieves zero-shot generalization across heterogeneous sensors, requiring no cameras at inference, no manual solver routing, and no external certifiable solvers or ICP.

In summary, our key contributions are three-fold:

1. We introduce a cross-modal distillation paradigm that transfers visual knowledge from DINOv2 (Oquab et al. 2024) to a 3D student network via hyperspherical manifold alignment ($\mathcal{L}_{\text{sph}}$). By optimizing true angular distances rather than flat Euclidean metrics, our formulation preserves the geometry of the vision foundation model, preventing feature distortion and improving point-level matching accuracy on sparse, cross-sensor scans.
2. We present a joint optimization strategy combining an intra-scan contrastive registration prior ($\mathcal{L}_{\text{infonce}}$) with a soft $\mathrm{SE}(3)$ invariance constraint ($\mathcal{L}_{\text{eq}}$) to maintain robust registration under geometric transformations. Executed via a single stacked pseudo-batch forward pass, this mechanism minimizes descriptor drift across perspective shifts and forces the network to capture spatial topologies.
3. We consolidate the entire cross-sensor registration pipeline into a single, unified checkpoint. By integrating a density-aware point-dropout strategy and a lightweight built-in initializer, SPHERE handles diverse sensor modalities and bridges the beam-count gap during deployment. Without external certifiable solvers or ICP, our framework achieves leading cross-sensor recall on HeLiPR and competitive performance on KITTI and nuScenes.

Related Work

Cross-modal distillation SLidR (Sautier et al. 2022), ScaLR (Puy, Boulch, and Marlet 2024), and DITR (Puy et al. 2024) establish that frozen ViT features can be lifted onto LiDAR points and distilled into sparse 3D backbones. However, their training objectives primarily target downstream semantic segmentation rather than point-level correspondence quality. Prior CLIP-style lifting methods explore general 3D understanding tasks but do not optimize outdoor registration correspondence quality. VFM-Registration (Vödisch et al. 2025) represents a closely related configuration, utilizing DINOv2 features for outdoor registration by projecting RGB patches onto points at inference time. While this method preserves foundation model quality, the requirement for synchronized and calibrated cameras at test time violates the LiDAR-only deployment constraint common in automotive stacks. DINOReg (Song, Xu, and Yan 2024) fuses DINOv2 with geometric cues but restricts its validation to indoor RGB-D scenarios. In contrast, SPHERE executes cross-modal distillation once during training, deploys a camera-free LiDAR-only pipeline at inference, and evaluates performance on heterogeneous outdoor benchmarks.

LiDAR registration GeoTransformer (Qin et al. 2022), MAC (Zhang et al. 2023), CAST (Huang et al. 2024), PARE-Net (Yao et al. 2024), and UGP (Zeng et al. 2025) learn geometric or hybrid correspondences, but their representations are optimized for rotating multi-beam sensor patterns encountered during training. To address this in-distribution dependency, BUFFER-X (Seo et al. 2025) introduces adaptive voxelization via structural sphericity and patch-wise scale normalization, enabling zero-shot registration across diverse scenes without environment-specific manual parameter tuning. Concurrently, RAP (Pan et al. 2025) departs from standard matching pipelines by casting registration as a generative conditional flow matching problem. It utilizes an alternating-attention transformer to estimate a continuous point-wise velocity field that directly transports unposed scans into a canonical registered frame, enforcing rigidity constraints during flow sampling at test time. On the learning-free side, frameworks such as KISS-Matcher (Lim et al. 2025) optimize scalability by pairing Faster-PFH descriptors with a linear-complexity k-core graph pruning algorithm to accelerate outlier rejection. However, because these methods rely on purely geometric cues, they remain vulnerable when spatial configurations become ambiguous due to extreme beam sparsity or out-of-distribution sensor patterns. SPHERE operates as a complementary training methodology rather than an architectural competitor, injecting vision-derived semantic anchors on top of the 3D backbone to preserve matching consistency.

Feature geometry The efficacy of learning-based 3D registration hinges on the intrinsic geometric structure of the feature embedding space. Traditionally, hard structural equivariance is explicitly integrated into networks such as PARE-Net (Yao et al. 2024), $\mathrm{SE}(3)$ transformers, and Vector Neurons (Deng et al. 2021) to preserve coordinate transformations within this geometric space. Conversely, in the metric learning domain, ArcFace and SphereFace demonstrate that enforcing angular margins on a hypersphere enhances feature discriminability for recognition tasks by shaping the topology of the embedding manifold. SPHERE bridges these paradigms by importing this hyperspherical metric directly into cross-modal distillation. Instead of treating normalized features as flat Euclidean vectors, it minimizes geodesic distance with a slack-tolerance threshold on the unit sphere, preventing the geometric distortion inherent in ambient space minimization. To maintain computational efficiency and remain backbone-agnostic, SPHERE employs soft invariance

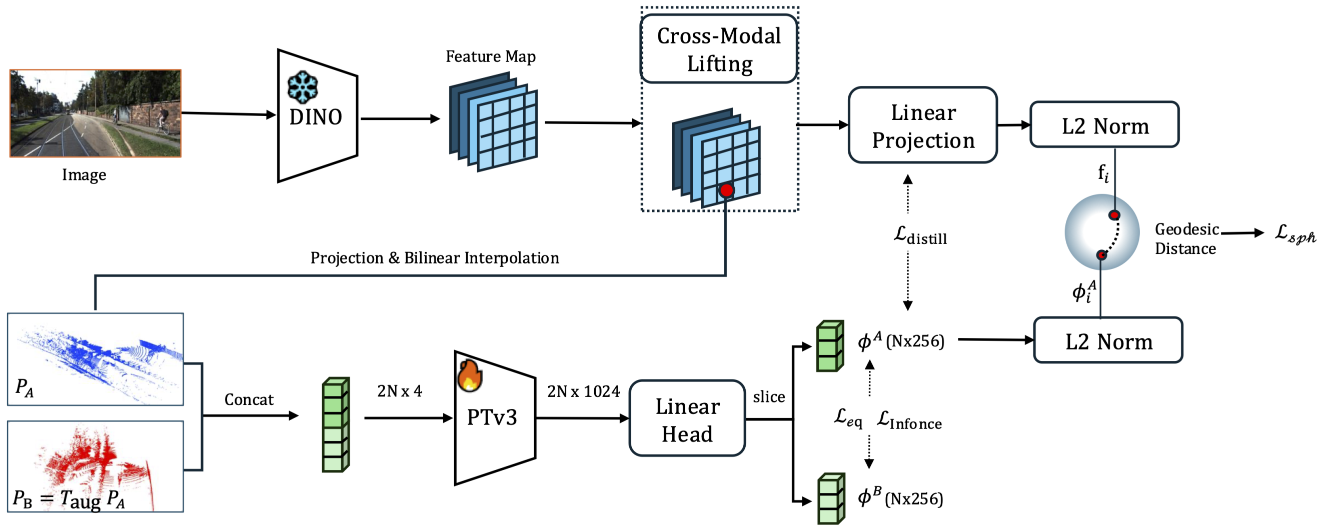

Figure 1: Overview of Stage-1 cross-modal distillation framework. (Left) Teacher branch: Dense 2D DINOv2 features are lifted onto visible 3D LiDAR points via projective geometry and bilinear interpolation. (Right) Student branch: A trainable PTv3 backbone processes original (P_A) and augmented (P_B) scans in a single stacked pseudo-batch pass. The representation is optimized jointly via cross-modal alignment ($\mathcal{L}_{\text{distill}}, \mathcal{L}_{\text{sph}}$) within the visibility mask $\mathcal{M}$ and self-supervised perspective consistency ($\mathcal{L}_{\text{infonce}}, \mathcal{L}_{\text{eq}}$).

via a paired fused forward pass rather than hard network constraints, while providing a path for future extensions via scalar descriptor fields.

Stage-1: Cross-Modal Distillation Pretraining

We establish a robust semantic foundation within the 3D student network by leveraging dense visual priors from a frozen 2D foundation network (Figure 1). In the teacher branch, DINOv2 patch features are lifted onto co-registered LiDAR coordinates via projective geometry and bilinear coordinate mapping. Concurrently, in the parallel student branch, the original point cloud P_A and a spatially augmented view P_B—with SE(3) applied to xyz coordinates only, as defined below—are jointly encoded through a stacked PTv3 forward pass. The primary objective of this pretraining phase is to bridge the inherent cross-modality domain gap between heterogeneous sensors while ensuring that the learned 3D geometric representations remain invariant to perspective shifts and spatial transformations.

Cross-Modal Feature Lifting To distill visual priors, we establish the spatial correspondence between the 2D image plane and the 3D LiDAR coordinate space. Given a synchronized frame with a raw LiDAR scan P and V calibrated camera views, each point $p_i = [x_i, y_i, z_i]^\top \in P$ (spatial coordinates only) is projected into the camera frame $v \in \{1, \ldots, V\}$ using the rigid extrinsic matrix $T_{\text{cam}\leftarrow\text{lidar}} \in \text{SE}(3)$:

$$\begin{bmatrix} P_{\text{cam}} \\ 1 \end{bmatrix} = T_{\text{cam}\leftarrow\text{lidar}} \begin{bmatrix} p_i \\ 1 \end{bmatrix}, \tag{1}$$

where $P_{\text{cam}} = [X_c, Y_c, Z_c]^\top$ represents the camera-centric coordinates. The continuous pixel coordinates (u, v) on the image plane are derived via perspective division:

$$u = \frac{K_v^{(0,0)} X_c + K_v^{(0,2)} Z_c}{Z_c}, \quad v = \frac{K_v^{(1,1)} Y_c + K_v^{(1,2)} Z_c}{Z_c}, \tag{2}$$

where $K_v \in \mathbb{R}^{3\times3}$ is the camera intrinsic matrix. To filter occlusions and boundary artifacts, p_i is mapped to view v if its depth $Z_c \in [0.5, 120]$ m and the projected coordinates satisfy $0 \le u < W_{\text{orig}}$ and $0 \le v < H_{\text{orig}}$, where W_{orig} and H_{orig} are the native image dimensions.

Concurrently, the RGB image is resized to 448×896 pixels and passed through the frozen DINOv2 network, outputting a feature map of size $1024 \times 32 \times 64$. To match the downsampled feature grid, the valid pixel coordinates (u, v) are normalized to $[-1, 1]^2$:

$$\hat{u} = \frac{2u}{W_{\text{orig}} - 1} - 1, \quad \hat{v} = \frac{2v}{H_{\text{orig}} - 1} - 1. \tag{3}$$

The normalized coordinates $(\hat{u}, \hat{v})$ are used to sample a 1024-dimensional visual feature vector from the feature map via bilinear interpolation. For points visible in multiple views, the extracted features are averaged to maintain viewpoint invariance, yielding the final teacher reference feature $f_i \in \mathbb{R}^{1024}$. Points invalid across all views are assigned a zero vector and excluded from the distillation target via a binary visibility mask $\mathcal{M}$.

Student Backbone Architecture We instantiate Point Transformer V3 (PTv3) (Wu et al. 2024) as our 3D student backbone. The input point cloud $P \in \mathbb{R}^{N\times4}$ is voxelized at a 0.1 m resolution and processed using encoder channel dimensions of $(64, 128, 256, 512, 1024)$ and decoder depths of $[3, 3, 12, 3]$. Spatial serialization paths follow the standard PTv3 sequence to manage attention patterns across

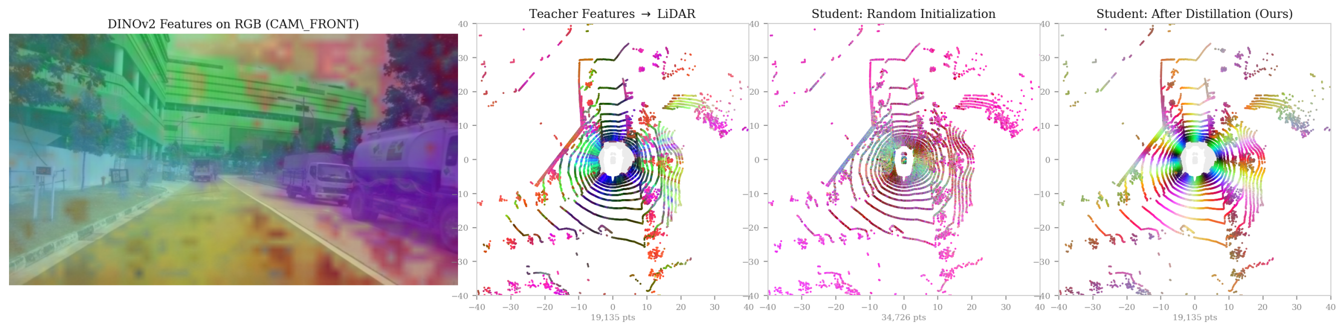

Figure 2: **Qualitative verification of cross-modal feature distillation via PCA color-space mapping.** (a) 2D DINOv2 features overlaid on the RGB frame (`CAM_FRONT`). (b) Lifted teacher features projected onto the 3D LiDAR space, where gray points indicate occlusion-masked elements outside $\mathcal{M}$. (c) Features from a randomly initialized student network. (d) Learned semantic feature topologies from the distilled student network.

varying point densities. The final decoder layer outputs a 1024-dimensional geometric feature map, which a bias-free linear head projects directly to a compact descriptor space, yielding $\phi_i \in \mathbb{R}^{256}$.

Stacked Dual-View Forward Pass To enforce coordinate transformation consistency without rigid structural constraints, we employ a soft invariance mechanism via pseudo-batch stacking. Given voxelized points P, we sample a random rigid transformation $T_{\text{aug}} = (R, t) \in \mathrm{SE}(3)$ independently per batch element, where the rotation R uses an angle $\theta \sim \mathcal{U}[-15°, +15°]$ along a normalized axis and the translation is $t \sim \mathcal{U}[-2, 2]^3$ m. T_{aug} is applied to spatial coordinates only: for each point $p_i = [x_i, y_i, z_i, I_i]^\top \in P$, the augmented counterpart is $p_i^{(B)} = [R[x_i, y_i, z_i]^\top + t;\ I_i]^\top$, leaving intensity unchanged. To maintain consistent batch normalization statistics (Ioffe and Szegedy 2015) and reduce overhead, the original scan $P_A = P$ and the perturbed scan P_B are concatenated along the batch dimension:

$$P_{\text{stacked}} = \text{concat}(P_A,\ P_B). \tag{4}$$

The student network processes P_{stacked} in a single pass to output a stacked descriptor matrix Φ, which is sliced into individual view descriptors:

$$\phi^{(A)} = \Phi_{0:N}, \quad \phi^{(B)} = \Phi_{N:2N}, \tag{5}$$

where N is the number of points per scan.

For self-supervision, we sample $M = 1024$ index-aligned anchor points and apply a symmetric InfoNCE loss (Oord, Li, and Vinyals 2018):

$$\begin{aligned}\mathcal{L}_{\text{infonce}} = \frac{1}{2}\Bigg(&\text{CE}\left(\frac{\tilde{\Phi}^{(A)}(\tilde{\Phi}^{(B)})^\top}{\tau}, \mathbf{y}\right)\\ &+ \text{CE}\left(\frac{\tilde{\Phi}^{(B)}(\tilde{\Phi}^{(A)})^\top}{\tau}, \mathbf{y}\right)\Bigg),\end{aligned} \tag{6}$$

where $\tilde{\Phi}^{(A)}, \tilde{\Phi}^{(B)} \in \mathbb{R}^{M \times d}$ are L_2-normalized sampled rows, $\tau = 0.07$ is the temperature, and $\mathbf{y} = (0, 1, \ldots, M-1)$ is the target index.

We minimize viewpoint drift by enforcing a soft $\mathrm{SE}(3)$ invariance constraint between index-aligned pairs:

$$\mathcal{L}_{\text{eq}} = \frac{1}{N}\sum_{i=1}^{N}\left(1 - \left\langle \tilde{\phi}_i^{(A)}, \tilde{\phi}_i^{(B)} \right\rangle\right), \tag{7}$$

where $\tilde{\phi}_i^{(A)}$ and $\tilde{\phi}_i^{(B)}$ are L_2-normalized descriptors. This stacked design preserves point ordering to guarantee valid index alignment. Without these objectives, the network processes only P_A and disables both $\mathcal{L}_{\text{infonce}}$ and $\mathcal{L}_{\text{eq}}$.

Hyperspherical Cross-Modal Alignment To transfer visual semantic priors, we map each lifted teacher feature $f_i \in \mathbb{R}^{1024}$ into the student embedding space via a bias-free linear projector $W_{\text{proj}} \in \mathbb{R}^{256 \times 1024}$, yielding $W_{\text{proj}} f_i \in \mathbb{R}^{256}$. The primary cross-modal objective combines cosine similarity and mean squared error (MSE) over the visibility mask $\mathcal{M}$:

$$\begin{aligned}\mathcal{L}_{\text{distill}} = \frac{1}{|\mathcal{M}|}\sum_{i \in \mathcal{M}}&\left(1 - \left\langle \tilde{\phi}_i^{(A)}, \tilde{f}_i \right\rangle\right)\\ &+ \lambda_{\text{mse}}\,\text{MSE}\big(\phi^{(A)}[\mathcal{M}],\ W_{\text{proj}} f[\mathcal{M}]\big),\end{aligned} \tag{8}$$

where $\tilde{f}_i = W_{\text{proj}} f_i / \|W_{\text{proj}} f_i\|_2$ and $\tilde{\phi}_i^{(A)} = \phi_i^{(A)} / \|\phi_i^{(A)}\|_2$ are the L_2-normalized projected teacher and student descriptors, respectively, and $\lambda_{\text{mse}} = 0.1$ balances the two terms.

To preserve the feature geometry of DINOv2 (Oquab et al. 2024) on the unit hypersphere $\mathbb{S}^{d-1}$ ($d = 256$), we incorporate a geodesic refinement loss with a slack-tolerance threshold $m = 0.2$ rad inspired by hyperspherical face recognition frameworks (Liu et al. 2017; Deng et al. 2019):

$$\mathcal{L}_{\text{sph}} = \frac{1}{|\mathcal{M}|}\sum_{i \in \mathcal{M}} \theta_i + \frac{1}{|\mathcal{M}|}\sum_{i \in \mathcal{M}} \max(0,\ \theta_i - m)^2, \tag{9}$$

where $\theta_i = \arccos\Big(\Big\langle \tilde{\phi}_i^{(A)}, \tilde{f}_i \Big\rangle\Big)$ is the geodesic angle on the hypersphere. Unlike $\mathcal{L}_{\text{distill}}$, $\mathcal{L}_{\text{sph}}$ directly optimizes angular alignment on the manifold rather than a linear cosine proxy, applying a squared hinge penalty when θ_i exceeds the slack-tolerance threshold m to sharpen sparse-scan correspondences.

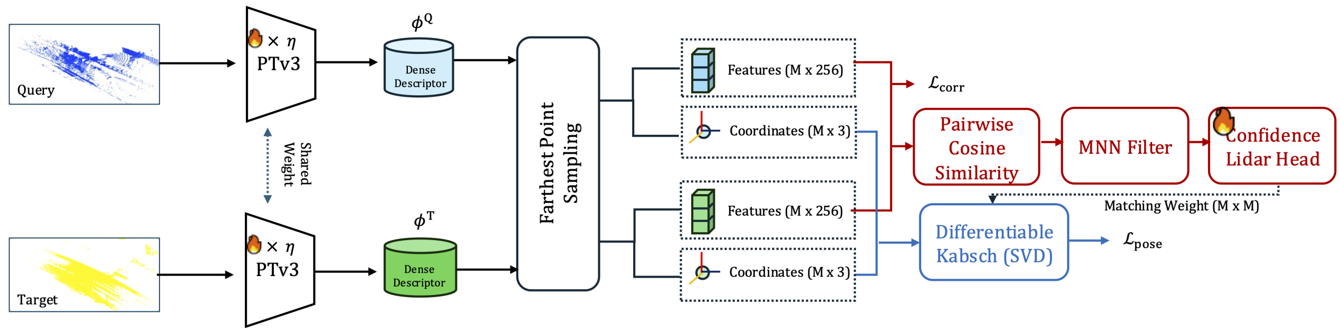

Figure 3: Overview of the Stage-2 pairwise registration fine-tuning pipeline. A weight-shared PTv3 backbone (Wu et al. 2024) extracts descriptors from query and target LiDAR scans. Farthest Point Sampling selects representative superpoints, which are matched via mutual nearest neighbors and weighted by a confidence head. A differentiable Kabsch solver (Kabsch 1976) utilizes these weighted correspondences to estimate the relative pose.

Figure 2 qualitatively compares an untrained student backbone (Fig. 2(c)) with the distilled network (Fig. 2(d)). The cross-modal distillation objectives guide the 3D feature representation to match the semantically consistent structure of the frozen 2D foundation teacher.

The total multi-task Stage-1 objective is formulated as:

$$\mathcal{L}_{\text{stage1}} = w_d \mathcal{L}_{\text{distill}} + w_r \mathcal{L}_{\text{infonce}} + w_{\text{sph}} \mathcal{L}_{\text{sph}} + w_{\text{eq}} \mathcal{L}_{\text{eq}}, \quad (10)$$

Stage-2: Pairwise Registration Fine-Tuning

Figure 3 summarizes the Stage-2 pairwise registration fine-tuning pipeline. Following Stage-1 pretraining, the 2D teacher network is removed, and the pretrained PTv3 backbone is fine-tuned on unposed LiDAR pairs $\{P_q, P_t\}$ within a weight-shared Siamese configuration. Dense descriptors from both scans are subsampled into superpoints via Farthest Point Sampling (FPS), matched via mutual nearest neighbors (MNN) with confidence weighting, and passed to a differentiable Kabsch solver for pose regression. This stage operates entirely without camera inputs or cross-modal projection. To adapt the geometric representations while preserving the distilled semantic priors, we employ a differential learning rate strategy alongside a newly appended registration head. A reduced learning rate multiplier is applied to the backbone layers, preventing catastrophic forgetting of Stage-1 knowledge. The joint objective minimizes a cross-scan contrastive matching loss $\mathcal{L}_{\text{corr}}$ and a weighted pose loss $\mathcal{L}_{\text{pose}}$.

Pairwise Superpoint Generation To reduce computational complexity, we downsample the dense scans into a sparse set of superpoints. Given the dense descriptors $\Phi_q, \Phi_t \in \mathbb{R}^{N\times 256}$, we apply Farthest Point Sampling (FPS) in the coordinate space to select $M = 512$ representative points. This yields superpoint coordinates $S_q, S_t \in \mathbb{R}^{M\times 3}$ and their corresponding feature vectors $G_q, G_t \in \mathbb{R}^{M\times 256}$, which serve as the primitives for correspondence estimation.

Correspondence Estimation and Matching Head We establish mutual nearest-neighbor (MNN) correspondences by selecting the top-K pairs based on the pairwise cosine similarity matrix between G_q and G_t. A two-layer confidence MLP (Linear→GELU→Linear, hidden dimension 64) processes the concatenated features of each candidate pair to predict a confidence weight $w_i \in [0, 1]$. The final correspondence score is formulated as $s_i = \text{ReLU}(\text{sim}_i) \cdot w_i$, which weights the inputs for the downstream Kabsch solver. The confidence MLP is optimized solely via $\mathcal{L}_{\text{pose}}$ gradients, requiring no auxiliary inlier supervision.

Registration Optimization Objectives The Stage-2 fine-tuning is driven by a joint objective combining a cross-scan contrastive loss $\mathcal{L}_{\text{corr}}$ and a weighted Kabsch pose loss $\mathcal{L}_{\text{pose}}$. We compute $\mathcal{L}_{\text{corr}}$ using an InfoNCE formulation over the superpoint features G_q and G_t. Concurrently, the relative transformation error is minimized via the Frobenius norm of the pose discrepancy:

$$\mathcal{L}_{\text{pose}} = \|T_{\text{gt}} - \text{Kabsch}(S_q, S_t, s)\|_F^2, \quad (11)$$

where s represents the final correspondence scores acting as alignment weights. This formulation forces the network to assign higher weights to geometrically stable anchors.

Point Dropout Augmentation To bridge the density and topology gap between dense training scans (KITTI Velodyne HDL-64E, $\approx$120k points; nuScenes 32-beam, $\approx$30k points) and sparse deployment environments (Velodyne VLP-16, $\approx$8k points), we apply stochastic point dropout during Stage-2 fine-tuning. HeLiPR is excluded from all training stages, so no Ouster-128 scans are seen during optimization. For each training step, the source and target point clouds are independently downsampled with probability $p_{\text{apply}} = 0.5$:

1. Sample a keep ratio uniformly: $\rho \sim \mathcal{U}[1 - d_{\max},\ 1]$ with maximum dropout strength $d_{\max} = 0.75$.
2. Retain a random subset of $\max(n_{\min}, \lfloor \rho N \rfloor)$ points drawn without replacement, where N is the input cloud size and $n_{\min} = 500$ ensures minimum spatial coverage.

This augmentation forces the student backbone to extract density-invariant geometric features; its effect on cross-sensor recall is quantified in Sec. .

Cross-Sensor Dataset Mixing The Stage-2 fine-tuning configuration incorporates training pairs from KITTI and nuScenes to enhance cross-sensor robustness. Exposing the backbone and confidence MLP to heterogeneous sensor patterns during fine-tuning improves cross-sensor generalization, as the correspondence head learns to produce robust confidence scores across varying point densities. To ensure balanced gradient updates and prevent a single dominant source from biasing the joint representation, we apply data source balancing to equalize the pair counts between the datasets during each epoch, stabilizing the joint embedding manifold.

Stage-2 Training Objective The joint Stage-2 objective optimizes both cross-scan correspondence quality and pose accuracy:

$$\mathcal{L}_{\text{stage2}} = w_c \mathcal{L}_{\text{corr}} + w_p \mathcal{L}_{\text{pose}}, \tag{12}$$

where $\mathcal{L}_{\text{corr}}$ is the cross-scan correspondence InfoNCE loss adapted from Equation (6), and $\mathcal{L}_{\text{pose}}$ is the weighted Kabsch pose loss from Equation (11). The loss weights are set to $w_c = 1.0$ and $w_p = 0.5$. To focus the gradient on high-confidence regions during dropout-augmented training, $\mathcal{L}_{\text{pose}}$ is computed using the top-128 correspondences by score. The confidence MLP receives supervision directly from the gradients of $\mathcal{L}_{\text{pose}}$, bypassing the need for explicit inlier labels.

Inference

At deployment, SPHERE operates camera-free on unposed LiDAR pairs $\{P_q, P_t\}$, without multi-modal synchronization or external certifiable solvers (TEASER, Open3D) or ICP. The same weight-shared PTv3 backbone and matching head from Stage 2 are reused; only the frozen 2D teacher is removed.

Descriptor Extraction and Correspondences Both scans are encoded into dense descriptors and downsampled to M=512 superpoints via farthest point sampling. Mutual nearest neighbors on superpoint features yield candidate pairs; a confidence MLP assigns weights $w_i \in [0, 1]$, forming scores $s_i = \text{ReLU}(\text{sim}_i) \cdot w_i$ identical to Stage 2 training. The top-K=128 pairs by s_i are retained for pose estimation.

Built-in Mini-RANSAC Initializer Plain weighted Kabsch on all correspondences is fragile under sparse cross-sensor inputs, so SPHERE first runs a lightweight 3-point mini-RANSAC over the top-128 pairs. In each iteration, a minimal sample proposes a pose; the hypothesis with the largest inlier count under a 0.6 m threshold is kept. We use N_{ransac}=2000 for large-displacement KITTI/nuScenes pairs and N_{ransac}=500 for short-range HeLiPR pairs. The best hypothesis seeds the subsequent refinement stage.

Local Geometric Refinement (LGR) Starting from the RANSAC pose, SPHERE runs N_{lgr}=50 LGR iterations. In iteration t, source superpoints are warped by the current estimate, and correspondence weights are updated as

$$w_k^{(t)} = s_k \cdot \mathbb{1}\{\|T^{(t-1)} s_k - t_k\|_2 < \sigma_d\}, \tag{13}$$

with σ_d=0.6 m, before re-estimating the pose via weighted Kabsch. The loop terminates when $\|T_{\text{new}} - T_{\text{old}}\|_F < 10^{-4}$. This built-in initializer and LGR pipeline is shared across all benchmarks; ablations in Sec. quantify its contribution relative to weighted Kabsch alone and external TEASER.

Experiments

Experimental Setup

Evaluation Paradigm We evaluate SPHERE using a single checkpoint and a unified inference recipe across all benchmarks, without per-sensor fine-tuning, specialized heads, or test-time synchronization. All baselines are re-evaluated on identical pair lists under the same point-cap constraints, without ICP; SPHERE additionally uses a built-in mini-RANSAC initializer and LGR rather than external certifiable solvers. Full training hyperparameters and benchmark protocols are provided in Appendix .

Evaluation Metrics The primary evaluation metric is strict registration success rate (SR@0.5 m/1°), complemented by relaxed thresholds (1 m/2° and 2 m/5°) and median translation and rotation errors. Alignment errors are computed using the synchronized GNSS/INS ground truth as the reference.

Evaluation Benchmarks

KITTI Odometry We evaluate on odometry sequences 08, 09, and 10 (Geiger, Lenz, and Urtasun 2012), comprising 555 consecutive-frame pairs from a 64-beam Velodyne HDL-64E. SPHERE is trained on sequences 00–07 in Stages 1–2; evaluation uses the disjoint held-out test split. All methods share the same fixed pair list.

nuScenes We sample 500 key-frame pairs with a 1–5 m trajectory gap from nuScenes validation scenes (Caesar et al. 2020) (32-beam LiDAR). SPHERE uses nuScenes training scenes in Stages 1–2 alongside KITTI; all evaluation pairs are drawn from held-out validation scenes only. All methods share the same fixed pair list.

HeLiPR Cross-Sensor To evaluate zero-shot cross-sensor generalization, we use held-out KAIST05, DCC05, and RIVER05 from HeLiPR (Kim et al. 2024). Reference maps are built from a 128-beam Ouster LiDAR; query streams use Velodyne VLP-16, Livox Avia, Aeva FMCW, and in-distribution Ouster. We pair 50 predefined query frames per sequence with the closest map frame within < 5 m, yielding 600 pairs ($3 \times 50 \times 4$ sensors). HeLiPR is excluded from all SPHERE training stages; all methods share the identical 600-pair list under unified xyz + intensity inputs and no ICP.

Main Results

Quantitative evaluation results across the three primary outdoor LiDAR benchmarks are summarized in Table 2, Table 1, and Table 3. SPHERE inference uses a built-in mini-RANSAC initializer and LGR without external certifiable solvers or ICP.

Our quantitative evaluation highlights three primary empirical insights regarding the representation quality and generalization limits of the framework.

Universal vs. Specialist Capabilities While high-density specialist models achieve high success rates on in-distribution KITTI data, they often fail to generalize when deployed zero-shot on unconventional HeLiPR scan patterns such as Livox Avia or Aeva FMCW. SPHERE remains competitive on KITTI (97.84% strict SR; CAST reaches 99.28%), achieves

Table 1: **Zero-shot cross-sensor generalization on the HeLiPR dataset.** All methods are evaluated zero-shot on HeLiPR without fine-tuning or adaptation. Bold font indicates the best performance in each column.

Method	Primary Metric: SR@0.5m/1° (%) ↑				Relaxed Metric: SR@1m/2° (%) ↑				Cross-Sensor Med. RTE (m) ↓
	Ouster (Same)	Velodyne 16-beam	Avia (Solid)	Aeva (FMCW)	Ouster (Same)	Velodyne 16-beam	Avia (Solid)	Aeva (FMCW)	
FPFH + TEASER	100.0	2.5	30.0	60.0	100.0	16.2	53.3	86.7	1.080
KISS-Matcher	100.0	6.2	31.2	1.2	100.0	36.2	70.0	45.0	0.515
RAP	100.0	93.8	12.5	11.2	100.0	93.8	46.2	43.8	1.061
CAST	100.0	0.0	0.0	0.0	100.0	0.0	0.0	0.0	6.115
GeoTransformer	100.0	60.0	0.0	0.0	100.0	80.0	0.0	0.0	0.164
UGP	100.0	18.8	45.0	15.0	100.0	78.8	56.2	37.5	0.470
PARE-Net	78.8	52.5	37.5	27.5	100.0	61.3	55.0	43.8	0.535
BUFFER-X	100.0	53.3	82.7	94.0	100.0	91.3	**100.0**	**99.3**	0.269
SPHERE (Ours)	**100.0**	**94.7**	**90.0**	**98.7**	**100.0**	**95.3**	91.3	98.7	**0.118**

Table 2: **Registration performance on the KITTI odometry benchmark.** SR@tm/r° denotes success rate at threshold (t, r); RTE and RRE are median translation and rotation errors. Bold font indicates the best performance in each column.

Method	Success Rate (%)			Median Error	
	SR@2m/5°	SR@1m/2°	SR@0.5m/1°	RTE (m) ↓	RRE (°) ↓
FPFH + TEASER	98.74	97.66	92.25	0.070	0.295
MAC	92.97	88.83	78.38	0.095	0.410
RAP	97.12	93.51	81.80	0.263	0.366
KISS-Matcher	97.12	87.75	66.31	0.140	0.723
UGP	99.82	99.64	97.66	0.060	0.189
PARE-Net	99.82	99.64	97.66	0.040	0.172
GeoTransformer	99.82	99.64	97.66	0.054	0.173
CAST	**100.00**	**100.00**	**99.28**	**0.023**	**0.125**
SPHERE	99.82	99.64	97.84	0.053	0.172

Table 3: **Registration on nuScenes.** SR@tm/r° denotes success rate at threshold (t, r); RTE and RRE are median translation and rotation errors. Bold font indicates the best performance in each column.

Method	Success Rate (%)			Median Error	
	SR@2m/5°	SR@1m/2°	SR@0.5m/1°	RTE (m) ↓	RRE (°) ↓
FPFH + TEASER	87.4	87.4	85.6	0.090	0.302
KISS-Matcher	78.4	77.2	67.8	0.124	0.523
GeoTransformer	99.4	98.6	97.0	0.080	**0.239**
BUFFER-X	92.6	90.6	89.6	0.113	0.294
SPHERE (Ours)	**100.0**	**100.0**	**99.2**	**0.078**	0.249

99.2% on nuScenes, and delivers the strongest HeLiPR cross-sensor strict recall (94.7% on Velodyne-16, 95.8% overall).

Cross-Sensor Robustness SPHERE yields substantial improvements over BUFFER-X across heterogeneous HeLiPR sensors at the strict threshold: +41.4 pp on Velodyne-16 (94.7% vs. 53.3%), +7.3 pp on Avia (90.0% vs. 82.7%), and +4.7 pp on Aeva (98.7% vs. 94.0%), with median translation error reduced from 0.269 m to 0.118 m. Point-dropout augmentation is critical on this benchmark (Appendix Table 6); the main HeLiPR result uses the built-in mRANSAC+LGR recipe (Appendix Table 5).

Table 4: Stage-1 loss ablation on HeLiPR 280-pair strict recall.

Configuration	Overall	Vel 16	Avia	Aeva
$\mathcal{L}_{\text{distill}}$	87.3	68.3	83.6	90.7
$+\mathcal{L}_{\text{eq}}$	89.5	76.0	87.2	89.8
$+\mathcal{L}_{\text{sph}}$	89.1	81.5	88.0	87.1
Full SPHERE (Ours)	**91.1**	**85.0**	**91.6**	**88.9**

Unified Built-in Inference All reported SPHERE metrics use one checkpoint with a built-in mini-RANSAC initializer (K=128) and 50 LGR iterations, without TEASER, Open3D, or ICP. Without this initializer, weighted Kabsch+LGR collapses to 5.5% on HeLiPR and 85.8% on nuScenes. The built-in recipe recovers most of this gap, trailing TEASER+LGR by 2.9 pp on HeLiPR (95.8% vs. 98.7%).

Ablation Studies

Stage-1 loss formulations. Table 4 ablates Stage-1 objectives on a 280-pair HeLiPR subset (70 pairs per sensor). $\mathcal{L}_{\text{sph}}$ primarily benefits the sparse Velodyne stream (+13.2 pp over $\mathcal{L}_{\text{distill}}$ alone), while $\mathcal{L}_{\text{eq}}$ improves overall stability (+2.2 pp). The full joint objective yields the best overall strict recall (+3.8 pp).

Robust initialization. Table 5 (Appendix) isolates the pose initializer under identical SPHERE correspondences. Without a robust hypothesis-and-verify stage, weighted Kabsch+LGR collapses to 5.5% overall recall on HeLiPR (Velodyne: 1.3%). Our built-in mini-RANSAC initializer reaches 95.8% overall (94.7% on Velodyne), recovering most of this gap and trailing external TEASER+LGR (98.7%) by only 2.9 pp; Open3D RANSAC reaches 96.2%. Further ablations of point-dropout strength and test-time VFM projection are in Appendix ; qualitative HeLiPR results are in Appendix .

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Training and Evaluation Details

Stage-1 pretraining. We pretrain on nuScenes and KITTI for 100k steps using AdamW (learning rate 10^{-4}, cosine decay) with the final layer of DINOv2 (ViT-L/14) as the distillation target. The Stage-1 loss weights in Equation (10) are $w_d = 1.0$, $w_r = 0.5$, $w_{\text{sph}} = 0.3$, and $w_{\text{eq}} = 0.3$. Images are resized to 448×896; DINOv2 outputs a $1024 \times 32 \times 64$ feature map. The PTv3 student voxelizes inputs at 0.1 m and projects decoder features to 256-d descriptors.

Stage-2 fine-tuning. We fine-tune for 20k steps on KITTI (sequences 00–07) and nuScenes training-scene pairs with per-epoch data-source balancing. KITTI evaluation uses sequences 08–10; nuScenes evaluation uses validation-scene pairs only. We apply a differential learning rate: trunk multiplier 0.05 ($2.5{\times}10^{-6}$) and head learning rate $5{\times}10^{-5}$. Point-dropout augmentation ($p_{\text{apply}}{=}0.5$, $d_{\max}{=}0.75$) is active during Stage-2, and $\mathcal{L}_{\text{pose}}$ uses the top-128 correspondences by score. HeLiPR is excluded from all training stages.

Evaluation protocols. **KITTI:** 555 consecutive-frame pairs from sequences 08–10 (64-beam Velodyne HDL-64E). **nuScenes:** 500 validation key-frame pairs with a 1–5 m gap (32-beam LiDAR). **HeLiPR:** held-out KAIST05, DCC05, and RIVER05; Ouster-128 reference maps with Velodyne VLP-16, Livox Avia, Aeva FMCW, and in-distribution Ouster queries (600 pairs total, $3 \times 50 \times 4$ sensors). All methods share identical pair lists under unified xyz + intensity inputs and no ICP. The primary metric is strict SR@0.5 m/1°, complemented by relaxed thresholds and median translation/rotation errors against GNSS/INS ground truth.

Additional Ablation Studies

Robust initialization. Table 5 isolates the pose initializer on HeLiPR (600 pairs, no ICP). All SPHERE variants share identical correspondences and are followed by LGR refinement.

Table 5: Solver ablation on HeLiPR cross-sensor.

Model	Solver	Overall	Vel.	Avia	Aeva
		SR@0.5m/1° (%) ↑			
Baseline					
BUFFER-X	RANSAC	82.5	53.3	82.7	94.0
No robust initializer					
SPHERE	wKabsch + LGR_{20}	5.5	1.3	0.0	0.0
External solver (upper bound)					
SPHERE	TEASER + LGR_{20}	98.2	96.7	96.7	99.3
SPHERE	TEASER + LGR_{50}	98.7	97.3	98.0	99.3
Built-in initializer (no external solver)					
SPHERE	O3D-RANSAC_{128} + LGR_{50}	96.2	97.3	89.3	98.0
SPHERE	**mRANSAC_{128} + LGR_{50} (Ours)**	**95.8**	**94.7**	**90.0**	**98.7**

Point-dropout augmentation. Table 6 isolates training-side dropout strength (evaluated with TEASER+LGR to separate solver effects). Dropout improves Velodyne strict recall from 91.3% to 97.3% at $d_{\max}{=}0.75$; smaller $d_{\max}$ recovers most of the cross-sensor gap at lower cost on Aeva.

Table 6: Point-dropout strength ablation on HeLiPR (600 pairs).

Dropout p	LGR iters	Overall	Vel.	Avia	Aeva	Med. RTE (m) ↓
		SR@0.5m/1° (%) ↑				
0 (none)	20	96.3	91.3	96.0	98.0	0.121
	50	96.8	91.3	97.3	98.7	0.118
0.50	20	98.2	95.3	97.3	**100.0**	**0.114**
	50	98.5	96.7	97.3	**100.0**	**0.114**
0.75	20	98.2	96.7	96.7	99.3	0.117
	50	**98.7**	**97.3**	**98.0**	99.3	0.118

Cross-modal distillation vs. test-time VFM projection. Table 7 compares distilled SPHERE against direct DINOv2-L projection at test time on a 100-frame KITTI pilot (not the full 555-pair benchmark). Direct projection achieves 24.0% strict SR with an identical downstream matcher, whereas the distilled student reaches 92.4% while remaining camera-free.

Table 7: Test-time VFM projection vs. distilled SPHERE (100 KITTI frames).

Descriptor Source	**Strict RR (%)** $\uparrow$	**Camera-Free?**
DINOv2-L projection (test-time)	24.0%	No
SPHERE (Ours)	**92.4%**	**Yes**

Qualitative Cross-Sensor Registration

Figure 4 visualizes representative HeLiPR cross-sensor pairs. Geometric matchers fail or degrade on non-repetitive Livox Avia and FMCW Aeva patterns, whereas SPHERE aligns wall structures and ground planes with low rotation error across all four scenarios.

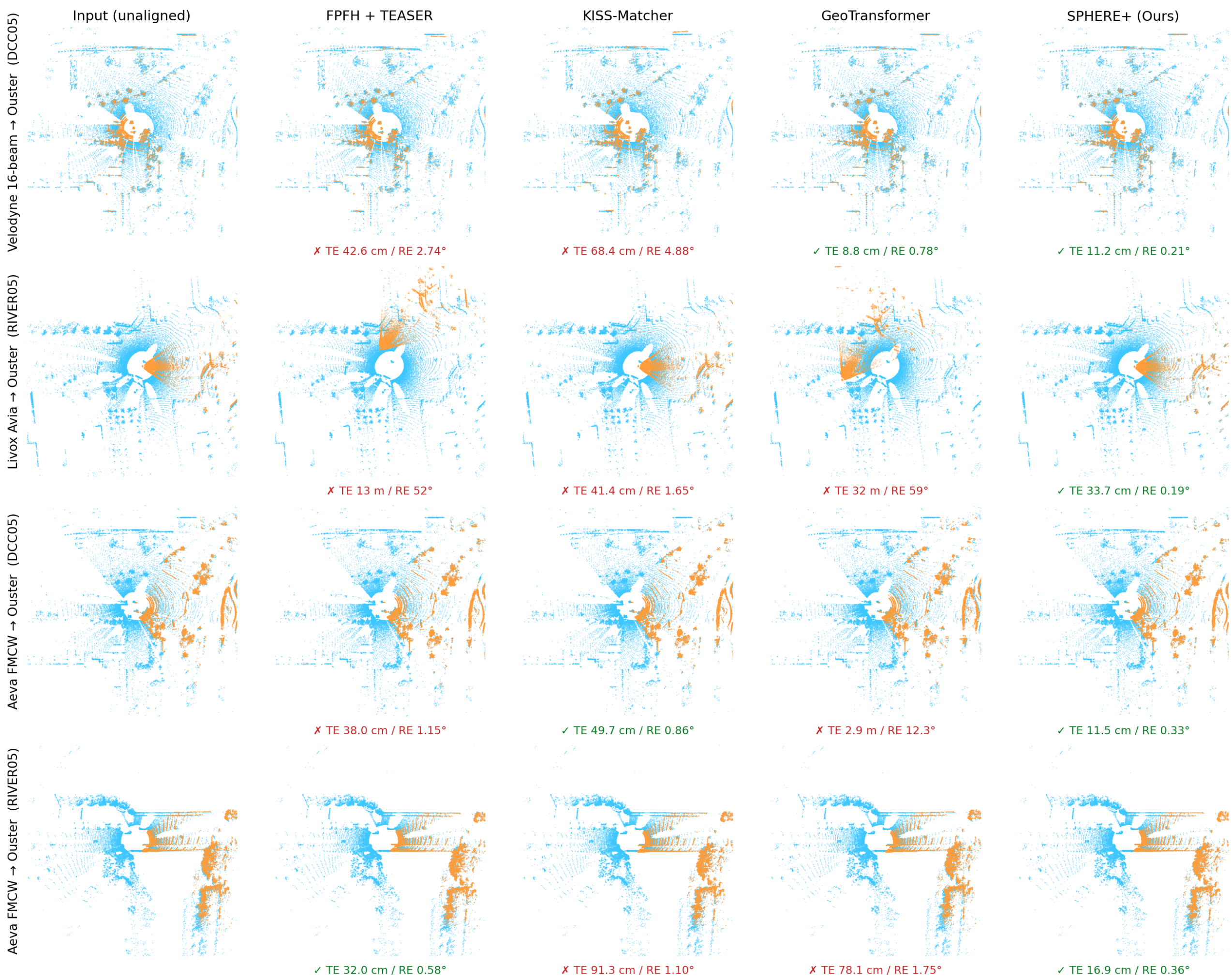

Figure 4: **Qualitative HeLiPR cross-sensor registration.** Each row shows a different query→map pairing against an Ouster-128 reference (blue); the transformed query is orange. Columns compare unaligned input, FPFH+TEASER, KISS-Matcher, GeoTransformer, and **SPHERE (Ours)**. Per-cell TE/RE report translation and rotation errors; ✓/✗ indicate success at SR@0.5 m/1°. SPHERE is the only method that succeeds on all four sensor pairings, including the challenging Livox Avia (solid-state) and Aeva FMCW streams.